

Machine Learning-Based COD Prediction Enhancement in Microalgae Cultivation with POME Using Imputation Techniques

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Abstract— Accurate COD prediction in microalgae-based Palm Oil Mill Effluent (POME) bioremediation systems is essential for effective process optimization and environmental protection. A critical challenge in real-time monitoring is the limitation of IoT sensors in directly measuring COD values, resulting in data availability restricted to periodic laboratory analysis which creates significant delays in process intervention. This study developed an IoT-based monitoring system integrated with imputation methods and machine learning algorithms to predict COD concentration during microalgae cultivation in POME media. To assess the impact of missing data handling on predictive performance, this research evaluated four machine learning algorithms on datasets prepared using four imputation strategies. Results demonstrated that MICE imputation combined with Random Forest achieved the best performance on per-minute data with R^2 of 0.99, RMSE of 2.97, and MAE of 0.86, significantly outperforming hourly aggregation which achieved R^2 of 0.95, RMSE of 32.66, and MAE of 12.22. Proper missing data handling at finer temporal granularity substantially enhanced predictive accuracy, enabling reliable real-time COD estimation without laboratory delays. The findings underscore the efficacy of integrating imputation methods with machine learning approaches to link real-time sensor data and laboratory-confirmed COD measurements in POME treatment facilities.

Keywords— COD prediction, data imputation, POME bioremediation, microalgae, IoT monitoring, machine learning

I. INTRODUCTION

The industrial processing of fresh fruit bunches in palm oil mills leads to the production of a liquid byproduct known as POME. Palm oil mills have a large production capacity and produce significant volumes of liquid by-products, known as POME [1]. POME is characterized by a high temperature, an acidic pH, and, most crucially, a high

organic matter content. The water sample exhibited remarkably elevated concentrations of organic matter, with biochemical oxygen demand (BOD) levels ranging from 8,200 to 35,400 milligrams per liter (mg/L), chemical oxygen demand (COD) ranging from 15,103 to 65,100 mg/L, and total solids (TS) content ranging from 16,580 to 94,106 mg/L [2]. If POME is discharged directly into the environment without adequate treatment, it can cause severe water pollution, threatening aquatic ecosystems and public health [3], [4].

Microalgae-based remediation represents an effective and sustainable approach for treating POME. Departing from established treatment methodologies, microalgae have been shown to facilitate the remediation of POME by metabolically uptaking its organic and inorganic constituents for biomass synthesis. This process has been demonstrated to reduce concentrations of chemical oxygen demand (COD), nitrogen, and phosphorus. [5]. This process not only serves to cleanse the waste, but it also produces algal biomass that has the potential to be processed into value-added products, such as biofuel or animal feed.

The efficacy of this treatment process is contingent upon the capacity to meticulously monitor and regulate bioreactor conditions. The implementation of IoT sensors facilitates real-time and continuous monitoring of critical parameters, including pH, temperature, dissolved oxygen levels, and light intensity [6]. The data collected from these sensors serves as the foundation for developing predictive models, particularly for predicting COD values. Accurate COD prediction is imperative because it enables operators to implement early intervention measures, optimize the treatment process, and ensure that the effluent produced adheres to environmental quality standards without the necessity of awaiting time-consuming laboratory analysis results [7].

Nonetheless, one major challenge in using IoT-based monitoring systems is making sure the data stays accurate and combining data from different sources. IoT sensors gather real-time data every minute, tracking things like pH, temperature, dissolved oxygen, and light intensity. However, COD measurements are restricted to periodic laboratory analysis, typically conducted daily or every few days due to the time-consuming nature of chemical analysis procedures. This creates a fundamental frequency mismatch between continuous sensor monitoring and intermittent laboratory measurements. Additionally, data transmitted from IoT sensors in the field is frequently incomplete or contains missing values [8]. The root causes of this phenomenon can be multifaceted, including but not limited to sensor malfunction, disruptions in data transmission networks, power outages, or periods designated for calibration and maintenance. Missing sensor data significantly reduces dataset quality, which in turn decreases the accuracy of the developed COD prediction model [9].

Filling in missing data with the right methods is important before starting any analysis. These methods can be as simple as using the last value seen, or more complex, like using imputation methods [8]. The choice of imputation method significantly impacts data distribution and model performance, yet comparative studies on IoT sensor data for microalgae-based POME treatment remain limited. This research addresses this gap by comparing multiple imputation methods on IoT sensor datasets from microalgae-based POME treatment systems to assess their impact on COD prediction accuracy [10], [11], [12].

The main contribution is identifying the optimal combination of imputation methods and machine learning algorithms to enhance COD prediction reliability and accuracy. This improved precision enables proactive optimization of waste treatment processes, increases operational efficiency, ensures compliance with environmental standards in POME treatment systems, and promotes sustainable palm oil industry practices.

II. LITERATUR REVIEW

A. AI and IoT System for Microalgae Cultivation in POME for COD Removal

POME is an aqueous byproduct of the extraction process used to obtain crude palm oil. It is characterized by its substantial organic load. The physicochemical profile of untreated POME is defined by its high pollution strength, which is evidenced by its Chemical Oxygen Demand (COD) levels, which range from 40,000 to 65,000 milligrams per liter. This is further compounded by elevated quantities of Biological Oxygen Demand (BOD), nitrogen, phosphorus, and suspended solids. Uncontrolled discharge of POME leads to eutrophication, hypoxia, ecosystem degradation, and biodiversity loss. The use of microalgae represents a promising and sustainable bioremediation approach. These photosynthetic microorganisms (5-several tens of micrometers) demonstrate remarkable metabolic tolerance to POME's extreme conditions and exceptional nutrient

accumulation capacity, making them ideal for agricultural wastewater phytoremediation.

COD reduction through microalgae treatment occurs via integrated heterotrophic and autotrophic metabolism. Microalgae assimilate complex organic compounds (polysaccharides, proteins, lipids) and dissolved minerals (ammonium, nitrate, phosphate) through active/passive transport, degrading them via intracellular enzymes (cellulase, protease, lipase). Through photosynthesis and the Calvin cycle, organic pollutants convert to microalgal biomass (amino acids, nucleotides, pigments), reducing COD dramatically to <1,000 mg/L and improving effluent clarity [13].

B. Data Imputation for Missing Value

Missing values are common in time-series datasets such as electronic nose sensor readings due to sensor malfunctions or data transmission errors [12]. Imputing missing data is essential for maintaining dataset accuracy. The methodological approach of this study involved the application of three data imputation strategies: K-Nearest Neighbors (KNN), linear interpolation, and Multiple Imputation by Chained Equations (MICE).

Linear interpolation is a technique that is used to estimate a missing value. This estimation is achieved by constructing a straight line. Eq. 1 show calculation for linier interpolation.

$$x_i^{imputed} = x_{i-k} + \frac{(x_{i+m} - x_{i-k})}{(i+m) - (i-k)} \times (i - (i-k)) \quad (1)$$

where $x_i^{imputed}$ is value at position i , x_{i-k} is the last observed non-missing value before i , and x_{i+m} is the next observed non-missing value after position i . This method provides realistic estimates for smooth data transitions.

K-Nearest Neighbors (KNN) imputation is a distance-based method that estimates missing values using the k most similar observations. Eq. 2 show calculation for KNN.

$$x_i^{imputed} = \frac{1}{k} \sum_{j=1}^k x_j^{(nearest)} \quad (2)$$

where $x_i^{imputed}$ is the imputed value, k is the number of nearest neighbors, and $x_j^{(nearest)}$ is the value nearest neighbor. Neighbors are identified using Euclidean distance. Eq. 3 show calculation for distance.

$$d(x_i, x_j) = \sqrt{\sum_{p=1}^n (x_{i,p} - x_{j,p})^2} \quad (3)$$

where $d(x_i, x_j)$ is the distance between observations i and j , and n is the total number of features. KNN is effective for multivariate datasets as it leverages feature relationships.

Multiple Imputation by Chained Equations (MICE) is an iterative method for filling missing data using a series of regression models. In this approach, each variable is imputed using the remaining variables as predictors. Eq. 4. show calculation for MICE.

$$\hat{x}_{i,j}^{(t)} = \beta_0 + \sum_{p \neq j}^n \beta_p x_{i,p}^{(t-1)} + \epsilon_i \quad (4)$$

where $\hat{x}_{i,j}^{(t)}$ is the imputed value for feature j in observation i at iteration t , β_0 is the intercept, β_p is the regression coefficient, and ϵ_i is the error term. MICE repeats this process iteratively for all features until convergence. This method maintains feature correlations and is particularly suitable for complex datasets like electronic nose sensor data [8], [14], [15].

C. Machine Learning for Prediction

Machine learning regression models have proven effective for predicting Chemical Oxygen Demand (COD) values in water quality assessment. The predictive capability for chemical oxygen demand (COD) was investigated using three regression models: The following algorithms will be examined: Random Forest (RF), Decision Tree (DT), and XGBoost. The evaluation of these models was conducted using the following metrics: the coefficient of determination, the root mean square error (RMSE), and the mean absolute error (MAE). The calculation for RF is provided in Eq. 5.

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (5)$$

where $\hat{y}$ is the predicted COD, T is the trees, and $f_t(x)$ is the prediction from tree t . RF reduces overfitting by averaging multiple tree predictions.

A Decision Tree (DT) is a supervised learning algorithm that recursively partitions the feature space to minimize prediction error. Eq. 6 show calculation for DT [16], [17].

$$MSE_{node} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2 \quad (6)$$

where MSE_{node} is the mean squared error, y_i is the actual COD, $\bar{y}$ is the average COD, and n is the samples. Decision trees are interpretable and handle non-linear patterns but may overfit without proper pruning [18], [19].

Extreme Gradient Boosting (XGBoost) is a sophisticated algorithm that sequentially constructs weak learners to correct errors made by previous models. This method optimizes a regularized objective function. Eq. 7 show calculation for XGBOOST.

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (7)$$

where Obj is the objective function, $l(y_i, \hat{y}_i)$ is the loss function measuring, $\Omega(f_k)$ is the regularization term for tree k , and K is the trees. XGBoost handles feature interactions and non-linear relationships effectively for COD prediction [20].

III. PROPOSED METHOD

This study analyzes the processes of data acquisition and predictive modeling for chemical oxygen demand (COD) removal. Fig. 1 shown proposed method.

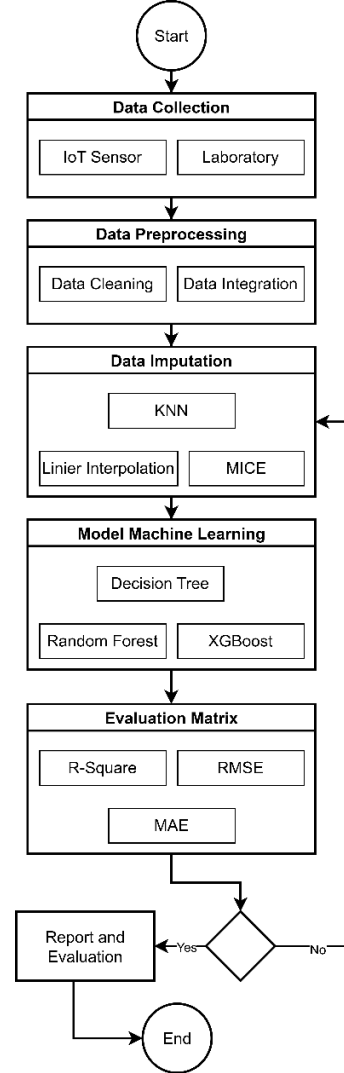


Fig 1. Proposed Method

A. Data Collection for IoT Sensor and Laboratory

The data collection system integrates real-time IoT sensor measurements with laboratory-based COD analysis in a hybrid monitoring approach. Internet of Things (IoT) sensors have been deployed to continuously monitor water quality parameters at five-second intervals. This process reduces the volume of data while maintaining temporal resolution. Table I presents an example dataset from the IoT sensor, COD measurements are obtained through conventional laboratory analysis, with a sampling frequency of once per day. As illustrated in Fig. 1 the COD data obtained was analyzed. This dual-frequency approach integrates the benefits of high-temporal-resolution sensor data for real-time monitoring with the precision of laboratory measurements for model calibration and validation purposes.

TABLE I. EXAMPLE DATASET FROM IOT SENSOR

Timestamp	Water Temp (°C)	Total Dissolved Solids (ppm)	Dissolved Oxygen (mg/L)	Water pH	Light Intensity (lux)	Atmospheric O ₂ (ppm)	Air Temp (°C)	Air Pressure (hPa)	Air Humidity (%)	Turbidity (NTU)
2025-08-14 15:38:00	32.63	1753.80	1.30	7.74	589.79	270334.77	39.75	1004.23	45.44	1230.00
2025-08-14 15:39:00	35.95	1851.18	1.70	8.17	590.14	270758.57	39.46	1004.20	45.81	35.67
2025-08-14 15:40:00	35.86	1849.58	1.69	8.24	588.20	270793.71	39.30	1004.18	46.03	35.50
2025-08-14 15:41:00	35.81	1860.69	1.73	8.38	587.64	271369.87	39.13	1004.21	46.26	33.67
2025-08-14 15:42:00	35.72	1860.24	1.73	8.30	586.39	270409.36	38.98	1004.19	46.54	32.67
...	...	...	...	...	...	...	...	...	...	...
2025-08-30 20:10:00	35.53	2357.81	3.81	10.09	1455.00	271019.52	37.37	1007.52	45.32	861.36
2025-08-30 20:11:00	35.52	2357.40	3.79	10.08	1454.79	270561.94	37.38	1007.55	45.32	900.26
2025-08-30 20:12:00	35.51	2359.02	3.79	10.10	1455.97	271901.69	37.41	1007.58	45.27	783.04
2025-08-30 20:13:00	35.50	2359.02	3.82	10.09	1457.50	272122.41	37.41	1007.59	45.26	852.99

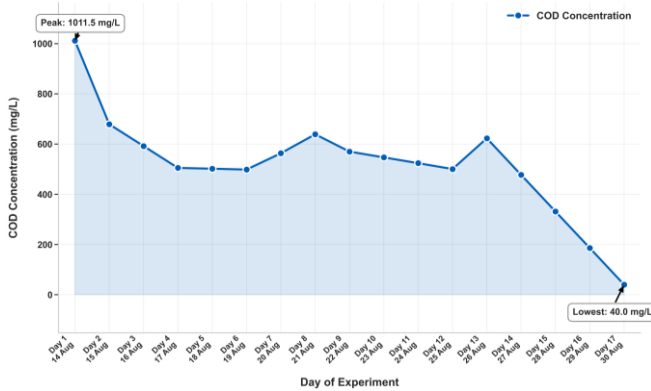


Fig 2. COD Data from Laboratory

B. Microalgae Cultivation Information

Microalgae cultivation was performed using a defined inoculum composition to optimize growth conditions. The cultivation medium comprised 500 mL of inoculum combined with 2000 mL of POME at a 20% concentration. To maintain optimal pH and supply a carbon source for photosynthesis, 7.5 grams of sodium bicarbonate was incorporated into the mixture. Continuous aeration was applied to facilitate oxygen and carbon dioxide exchange and to ensure uniform mixing of the culture. Light intensity was controlled at 1000 lux to provide sufficient photosynthetic active radiation for microalgae growth. The initial cultivation phase was maintained under these conditions for 24 hours to allow proper acclimatization and exponential growth initiation of the microalgae culture before the main experimental period commenced.

C. Data Imputation

A data imputation process was implemented to address the temporal discrepancy between continuous IoT sensor measurements and discrete daily COD measurements. A comparative analysis of three methodologies was conducted. The first method is known as K-Nearest Neighbors (KNN), while the second is linear interpolation. The third method is referred to as MICE. The objective of the study was to transform daily laboratory COD measurements into hourly and minute-level data points that were aligned with high-frequency IoT recordings. KNN imputation estimates missing values using similarity patterns from neighboring data points based on multiple sensor features. Linear interpolation is a statistical technique used to impute missing values by constructing a straight line. MICE employs iterative regression models that consider

multivariate relationships among all variables to predict missing values. By densifying laboratory COD data to align with the temporal resolution of IoT measurements, these techniques facilitated comprehensive analysis.

D. Model Machine Learning and Evaluation

Three tree-based machine learning algorithms were employed to develop models for estimating chemical oxygen demand (COD): Decision Tree, Random Forest, and XGBoost. Model evaluation employed three metrics: R-squared quantified the proportion of explained variance, Mean Squared Error (MSE) represented the average squared difference with greater emphasis on larger errors, and Mean Absolute Error (MAE) indicated the average absolute difference in the target variable.

IV. RESULT

A. Dataset Information

Data collection employed a combined approach of real-time IoT monitoring and periodic laboratory analysis. The IoT system continuously recorded process parameters every minute from August 14-22, 2025, generating 67,869 data rows with 98.3% completeness (11,314 of 11,507 expected minutes), averaging 6 readings per minute for measurement accuracy. Complementing this, COD samples were collected daily and analyzed using a HACH DR900 COD meter for laboratory validation. Data synchronization between IoT sensors and laboratory measurements was performed using timestamp alignment, with sensor readings averaged over 24-hour periods to match the daily laboratory sampling intervals.

B. Evaluation Result

Model performance was evaluated at two temporal scales: per-minute and hourly. For per-minute data, missing Chemical Oxygen Demand (COD) values were imputed at the sensor reading frequency to enable detailed real-time analysis. Conversely, the hourly dataset was generated by aggregating or averaging the per-minute sensor readings into one-hour intervals. Both datasets were divided using an 80-20 train-test split.

Table II presents evaluation results of imputation method and prediction model combinations on hourly data using R², RMSE, and MAE metrics. Baseline mean imputation yielded varied outcomes: Decision Tree achieved R² of 0.86 (RMSE: 54.49, MAE: 47.72), while XGBoost

reached the highest R^2 of 0.96 but with elevated RMSE of 56.24 and MAE of 10.58. Linear interpolation proved effective, particularly with Random Forest (R^2 : 0.95, RMSE: 46.18, MAE: 21.86) and LightGBM (R^2 : 0.94, RMSE: 48.45, MAE: 23.75). KNN imputation showed consistent results with R^2 ranging from 0.83 to 0.88, with Random Forest achieving R^2 of 0.88 (RMSE: 45.03, MAE: 16.80). MICE imputation demonstrated optimal overall performance: Random Forest achieved R^2 of 0.95 (RMSE: 32.66, MAE: 12.22), closely followed by LightGBM (R^2 : 0.95, RMSE: 32.65, MAE: 15.05). The MICE-Random Forest combination exhibited superior accuracy and consistency.

Table III presents per-minute data imputation evaluation results, showing more granular temporal resolution than hourly data. Baseline Mean imputation produced varied results: Decision Tree and XGBoost both achieved R^2 of 0.89 (RMSE: 10.64 and 8.38 respectively), while Random Forest showed R^2 of 0.81 with lowest RMSE of 4.27 and MAE of 0.99. Linear Interpolation demonstrated excellent performance with consistently high R^2 of 0.98-0.99 across all models, where XGBoost achieved best results (R^2 : 0.99, RMSE: 20.75, MAE: 7.12), followed by Random Forest (R^2 : 0.99, RMSE: 20.93, MAE: 6.51). KNN imputation showed outstanding performance with all models reaching R^2 of 0.99, where Random Forest delivered optimal combination (RMSE: 9.54, MAE: 2.14) and Decision Tree achieved RMSE of 12.74 and MAE of 1.75. MICE imputation emerged as superior with all models achieving R^2 of 0.99, where Random Forest demonstrated best overall performance with lowest RMSE of 2.97 and MAE of 0.86, followed by LightGBM (RMSE: 3.24, MAE: 1.36) and XGBoost (RMSE: 4.65, MAE: 1.25). The MICE-Random Forest combination proved optimal for per-minute data.

The MICE-Random Forest combination consistently exhibited superior performance across both temporal resolutions³. The efficacy of this approach is attributable to the strategic integration of MICE, which effectively captures intervariable correlations for missing value imputation, and Random Forest, which utilizes ensemble learning to model complex non-linear patterns in water quality data⁴. Critically, this integration was chosen because MICE addresses the core challenge of POME data namely, the scarcity of discrete daily laboratory data by generating validated high-resolution data, while Random Forest provides the optimal balance between high accuracy and efficient computational cost compared to more expensive and data-intensive deep learning alternatives. The combination's consistent excellence, particularly on per-minute data (with an MAE of 0.86), establishes new technical knowledge by proving this framework is the most robust and reliable approach for real-time COD monitoring in POME bioremediation systems. The scientific contribution lies not just in the integration, but in the empirical validation of a transformative level of precision that enables proactive operational intervention, moving beyond proof-of-concept to a technically validated

TABLE II. EVALUATION METRICS FOR PER HOURLY DATA IMPUTATION

Imputation Method	Model	R-Square	RMSE	MAE
Baseline (Mean)	Decision Tree	0.86	54.49	47.72
	Random Forest	0.79	56.18	19.59
	LightGBM	0.74	53.18	13.80
	XGBoost	0.86	56.24	19.58
Linear Interpolation	Decision Tree	0.81	86.52	31.32
	Random Forest	0.95	46.18	21.86
	LightGBM	0.94	48.45	23.75
	XGBoost	0.92	55.63	23.80
KNN	Decision Tree	0.83	53.73	19.93
	Random Forest	0.88	45.03	16.80
	LightGBM	0.86	47.73	24.34
	XGBoost	0.87	47.23	19.41
MICE	Decision Tree	0.93	38.76	15.01
	Random Forest	0.95	32.66	12.22
	LightGBM	0.95	32.65	15.05
	XGBoost	0.92	41.24	16.73

TABLE III. EVALUATION METRICS FOR PER-MINUTE DATA IMPUTATION

Imputation Method	Model	R-Square	RMSE	MAE
Baseline (Mean)	Decision Tree	0.89	10.64	0.93
	Random Forest	0.81	4.27	0.99
	LightGBM	0.69	3.10	1.41
	XGBoost	0.89	8.38	0.94
Linear Interpolation	Decision Tree	0.98	22.05	6.61
	Random Forest	0.99	20.93	6.51
	LightGBM	0.99	20.99	7.92
	XGBoost	0.99	20.75	7.12
KNN	Decision Tree	0.99	12.74	1.75
	Random Forest	0.99	9.54	2.14
	LightGBM	0.99	10.14	3.79
	XGBoost	0.99	10.72	3.78
MICE	Decision Tree	0.99	5.78	1.43
	Random Forest	0.99	2.97	0.86
	LightGBM	0.99	3.24	1.36
	XGBoost	0.99	4.65	1.25

V. CONCLUSION

This study evaluates different imputation methods and machine learning algorithms for predicting COD in microalgae-based POME treatment systems. It also addresses data integrity challenges in IoT-based monitoring. POME is characterized by elevated organic matter and high COD values, necessitating effective treatment to prevent environmental contamination. Although IoT sensors facilitate real-time monitoring of critical parameters, sensor data often contains missing values due to malfunctions,

network disruptions, or maintenance, which can compromise the reliability of prediction models. Conventional laboratory COD analysis is time-intensive and impedes process optimization. The findings indicate that robust imputation strategies are crucial for integrating continuous sensor monitoring with laboratory-validated measurements to achieve accurate real-time COD prediction.

The experimental results demonstrated that MICE imputation combined with Random Forest regression achieved the best performance on per-minute data with R^2 of 0.99, RMSE of 2.97, and MAE of 0.86, outperforming the hourly data results which achieved R^2 of 0.95, RMSE of 32.66, and MAE of 12.22. The superior performance on per-minute data indicates that higher temporal resolution captures more detailed variations in water quality parameters, enabling more precise COD prediction.

Future research should focus on developing deep learning architectures for time-series forecasting, implementing adaptive imputation strategies for varying missing data patterns, integrating automated control systems for dynamic bioreactor optimization, and conducting field validation across multiple POME treatment facilities to enhance industrial scalability and sustainable palm oil effluent management.

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